



Association Analysis I

Lecture 9

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Association Analysis I

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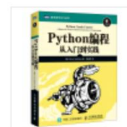
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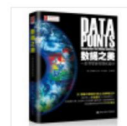
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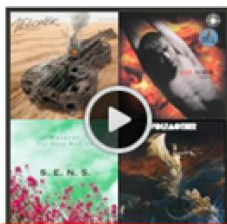
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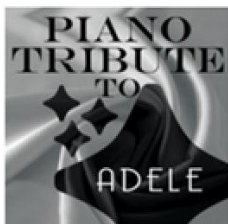
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Market-based Problem

- Purchase together: **Frequent items**
- $A \rightarrow B$: **Association rule**

It can be used to conduct:

- Cross selling
- Layout design
- Others

- Basic Concepts
 - Item, Confidence, Support
- Apriori Algorithm
 - Frequent Item generation
 - Rule generation
- Pattern Evaluation



Outline

- **Basic Concepts**
 - Item, Confidence, Support
- Apriori Algorithm
 - Frequent Item generation
 - Rule generation
- Pattern Evaluation

Association Rule Mining

- Given a set of transactions, find rules that will predict the occurrence of an item based on the occurrences of other items in the transaction

Market-Basket transactions

<i>TID</i>	<i>Items</i>
1	Bread, Milk
2	Bread, Diaper, Beer, Eggs
3	Milk, Diaper, Beer, Coke
4	Bread, Milk, Diaper, Beer
5	Bread, Milk, Diaper, Coke

Example of Association Rules

$\{\text{Diaper}\} \rightarrow \{\text{Beer}\},$
 $\{\text{Milk, Bread}\} \rightarrow \{\text{Eggs, Coke}\},$
 $\{\text{Beer, Bread}\} \rightarrow \{\text{Milk}\},$

Implication means co-occurrence,
not causality!

Definition: Frequent Itemset

- **Itemset**
 - A collection of one or more items
 - Example: {Milk, Bread, Diaper}
 - k-itemset
 - An itemset that contains k items
- **Support count (σ)**
 - Frequency of occurrence of an itemset
 - E.g. $\sigma(\{\text{Milk, Bread, Diaper}\}) = 2$
- **Support**
 - Fraction of transactions that contain an itemset
 - E.g. $s(\{\text{Milk, Bread, Diaper}\}) = 2/5$
- **Frequent Itemset**
 - An itemset whose support is greater than or equal to a *minsup* threshold

<i>TID</i>	<i>Items</i>
1	Bread, Milk
2	Bread, Diaper, Beer, Eggs
3	Milk, Diaper, Beer, Coke
4	Bread, Milk, Diaper, Beer
5	Bread, Milk, Diaper, Coke

Definition: Association Rule

- Association Rule
 - An implication expression of the form $X \rightarrow Y$, where X and Y are itemsets
 - Example:
 $\{\text{Milk, Diaper}\} \rightarrow \{\text{Beer}\}$

<i>TID</i>	<i>Items</i>
1	Bread, Milk
2	Bread, Diaper, Beer, Eggs
3	Milk, Diaper, Beer, Coke
4	Bread, Milk, Diaper, Beer
5	Bread, Milk, Diaper, Coke

- Rule Evaluation Metrics

- Support (s)
 - ◆ Fraction of transactions that contain both X and Y
- Confidence (c)
 - ◆ Measures how often items in Y appear in transactions that contain X

Example:

$\{\text{Milk, Diaper}\} \Rightarrow \text{Beer}$

$$s = \frac{\sigma(\text{Milk, Diaper, Beer})}{|T|} = \frac{2}{5} = 0.4$$

$$c = \frac{\sigma(\text{Milk, Diaper, Beer})}{\sigma(\text{Milk, Diaper})} = \frac{2}{3} = 0.67$$

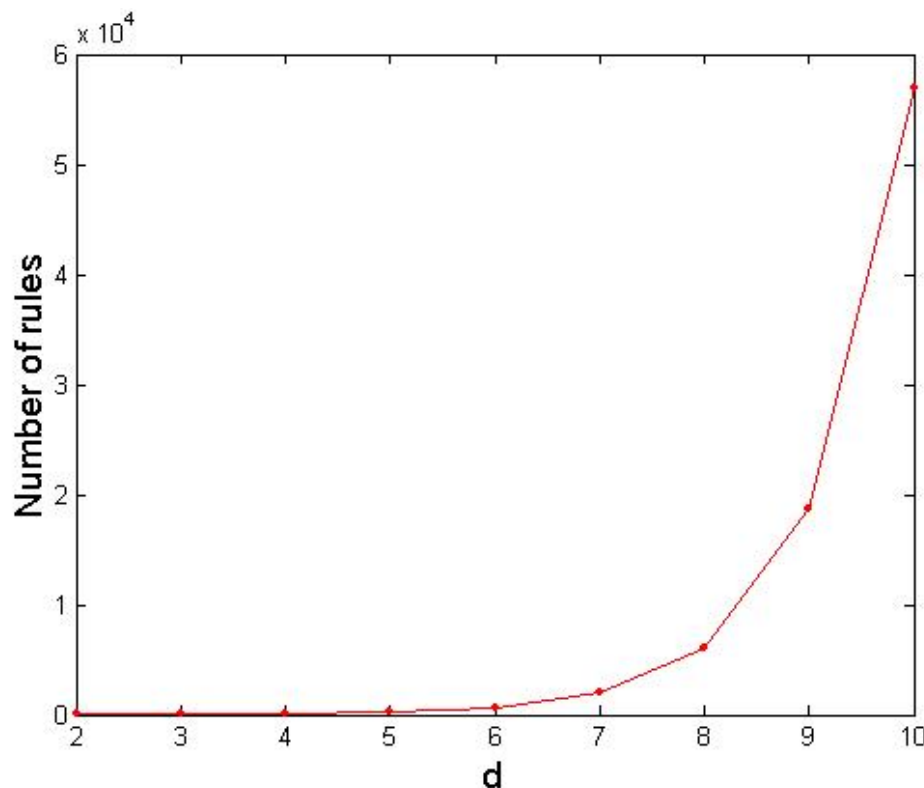
Association Rule Mining Task

- Given a set of transactions T , the goal of association rule mining is to find all rules having
 - support $\geq \textit{minsup}$ threshold
 - confidence $\geq \textit{minconf}$ threshold
- Brute-force approach:
 - List all possible association rules
 - Compute the support and confidence for each rule
 - Prune rules that fail the *minsup* and *minconf* thresholds

⇒ **Computationally prohibitive!**

Computational Complexity

- Given d unique items:
 - Total number of itemsets = $2^d - 1$
 - Total number of possible association rules:



$$R = \sum_{k=1}^{d-1} \left[\binom{d}{k} \times \sum_{j=1}^{d-k} \binom{d-k}{j} \right]$$
$$= 3^d - 2^{d+1} + 1$$

If $d=6$, $R = 602$ rules

- Basic Concepts
 - Item, Confidence, Support
- Apriori Algorithm
 - Frequent Item generation
 - Rule generation
- Pattern Evaluation

Mining Association Rules

<i>TID</i>	<i>Items</i>
1	Bread, Milk
2	Bread, Diaper, Beer, Eggs
3	Milk, Diaper, Beer, Coke
4	Bread, Milk, Diaper, Beer
5	Bread, Milk, Diaper, Coke

Example of Rules:

$\{\text{Milk, Diaper}\} \rightarrow \{\text{Beer}\}$ ($s=0.4, c=0.67$)
 $\{\text{Milk, Beer}\} \rightarrow \{\text{Diaper}\}$ ($s=0.4, c=1.0$)
 $\{\text{Diaper, Beer}\} \rightarrow \{\text{Milk}\}$ ($s=0.4, c=0.67$)
 $\{\text{Beer}\} \rightarrow \{\text{Milk, Diaper}\}$ ($s=0.4, c=0.67$)
 $\{\text{Diaper}\} \rightarrow \{\text{Milk, Beer}\}$ ($s=0.4, c=0.5$)
 $\{\text{Milk}\} \rightarrow \{\text{Diaper, Beer}\}$ ($s=0.4, c=0.5$)

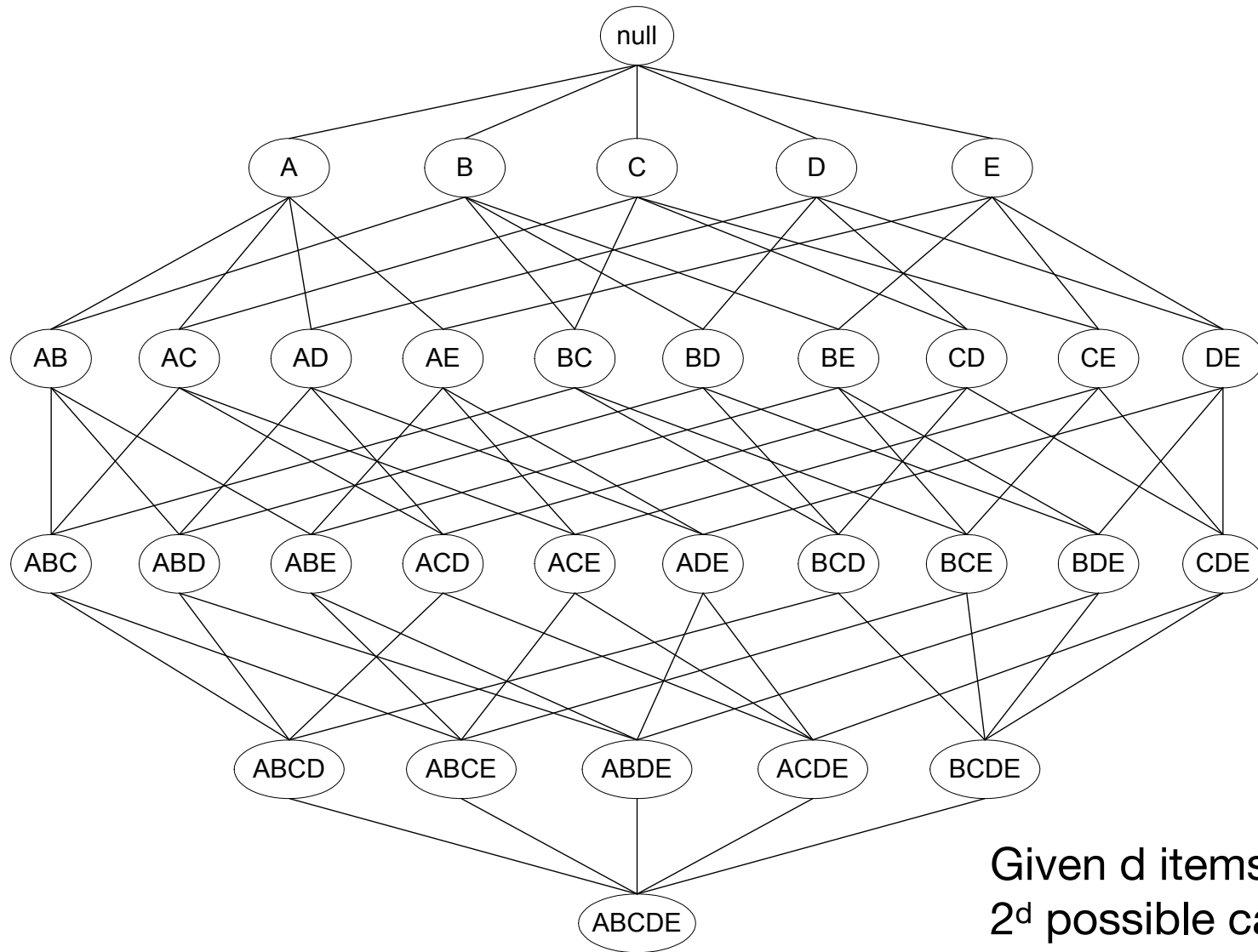
Observations:

- All the above rules are binary partitions of the same itemset:
 $\{\text{Milk, Diaper, Beer}\}$
- Rules originating from the same itemset have identical support but can have different confidence
- Thus, we may decouple the support and confidence requirements

Mining Association Rules

- Two-step approach:
 1. Frequent Itemset Generation
 - Generate all itemsets whose support $\geq$ minsup
 2. Rule Generation
 - Generate high confidence rules from each frequent itemset, where each rule is a binary partitioning of a frequent itemset
- Frequent itemset generation is still computationally expensive

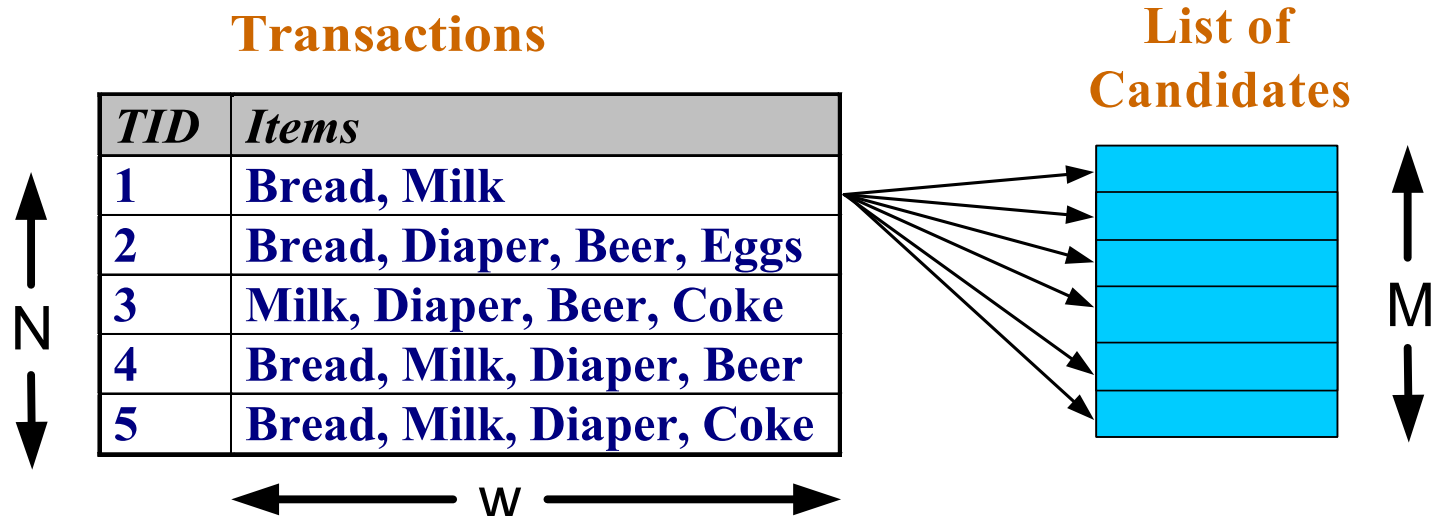
Frequent Itemset Generation



Given d items, there are 2^d possible candidate itemsets

Frequent Itemset Generation

- Brute-force approach:
 - Each itemset in the lattice is a **candidate** frequent itemset
 - Count the support of each candidate by scanning the database



- Match each transaction against every candidate
- Complexity $\sim O(NMw) \Rightarrow$ **Expensive since $M = 2^d - 1$!!!**

Frequent Itemset Generation Strategies

- Reduce the **number of candidates** (M)
 - Complete search: $M=2^d-1$
 - Use pruning techniques to reduce M
- Reduce the **number of transactions** (N)
 - Reduce size of N as the size of itemset increases
- Reduce the **number of comparisons** (NM)
 - Use efficient data structures to store the candidates or transactions
 - No need to match every candidate against every transaction

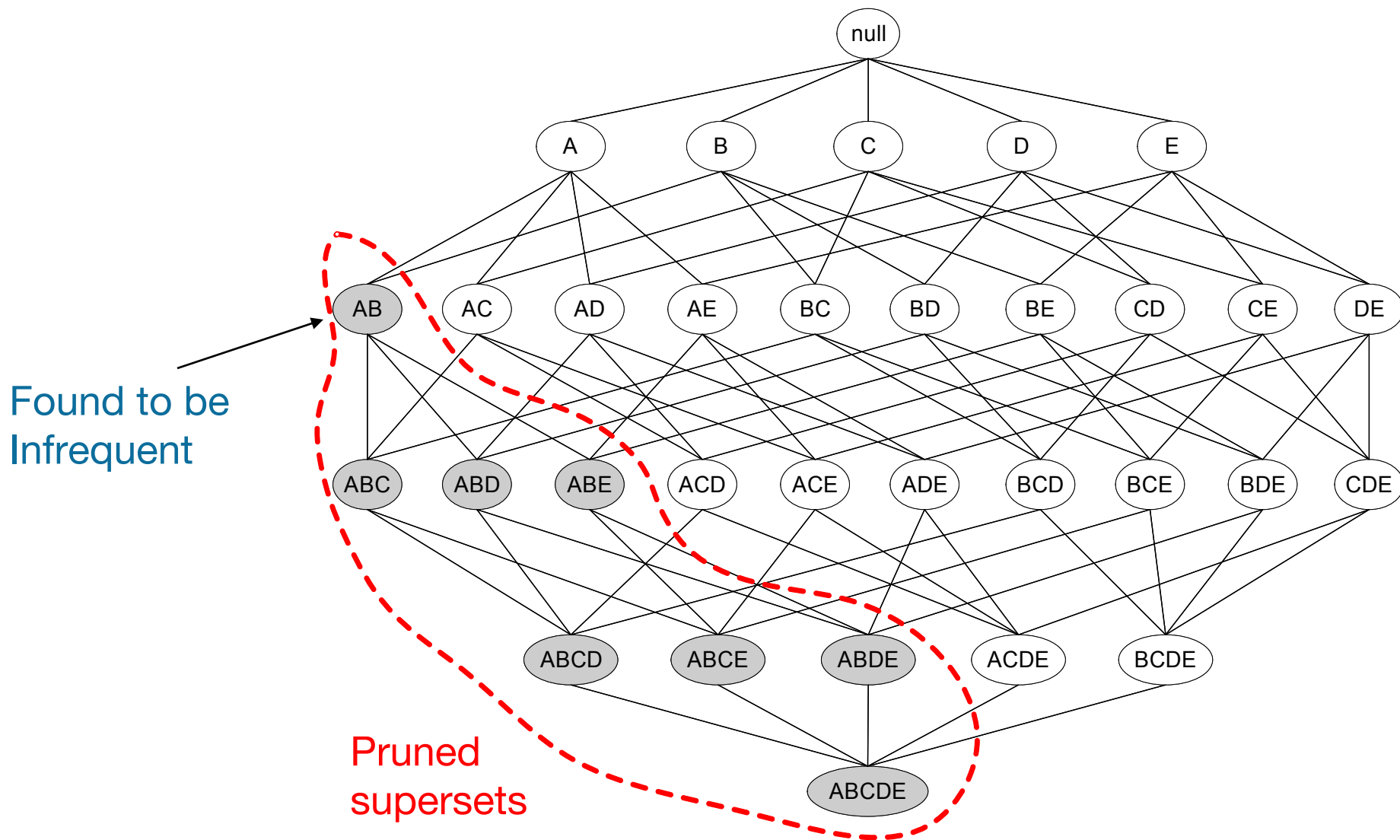
Reducing Number of Candidates

- **Apriori principle:**
 - If an itemset is frequent, then all of its subsets must also be frequent
- Apriori principle holds due to the following property of the support measure:

$$\forall X, Y : (X \subseteq Y) \Rightarrow s(X) \geq s(Y)$$

- Support of an itemset never exceeds the support of its subsets
- This is known as the **anti-monotone** property of support

Illustrating Apriori Principle



Apriori Algorithm

- Method:
 - Let $k=1$
 - Generate frequent itemsets of length 1
 - Repeat until no new frequent itemsets are identified
 - Generate length $(k+1)$ candidate itemsets from length k frequent itemsets
 - Prune candidate itemsets containing subsets of length k that are infrequent
 - Count the support of each candidate by scanning the DB
 - Eliminate candidates that are infrequent, leaving only those that are frequent

Illustrating Apriori Principle

Item	Count
Bread	4
Coke	2
Milk	4
Beer	3
Diaper	4
Eggs	1

Items (1-itemsets)

Minimum Support = 3



Itemset	Count
{Bread,Milk}	3
{Bread,Beer}	2
{Bread,Diaper}	3
{Milk,Beer}	2
{Milk,Diaper}	3
{Beer,Diaper}	3

Pairs (2-itemsets)

(No need to generate candidates involving Coke or Eggs)



Triplets (3-itemsets)

TID	Items
1	Bread, Milk
2	Bread, Diaper, Beer, Eggs
3	Milk, Diaper, Beer, Coke
4	Bread, Milk, Diaper, Beer
5	Bread, Milk, Diaper, Coke

Item set	Count
{Bread,Milk,Diaper}	3

If every subset is considered,
 $C_6^1 + C_6^2 + C_6^3 = 41$
 With support-based pruning,
 $6 + 6 + 1 = 13$

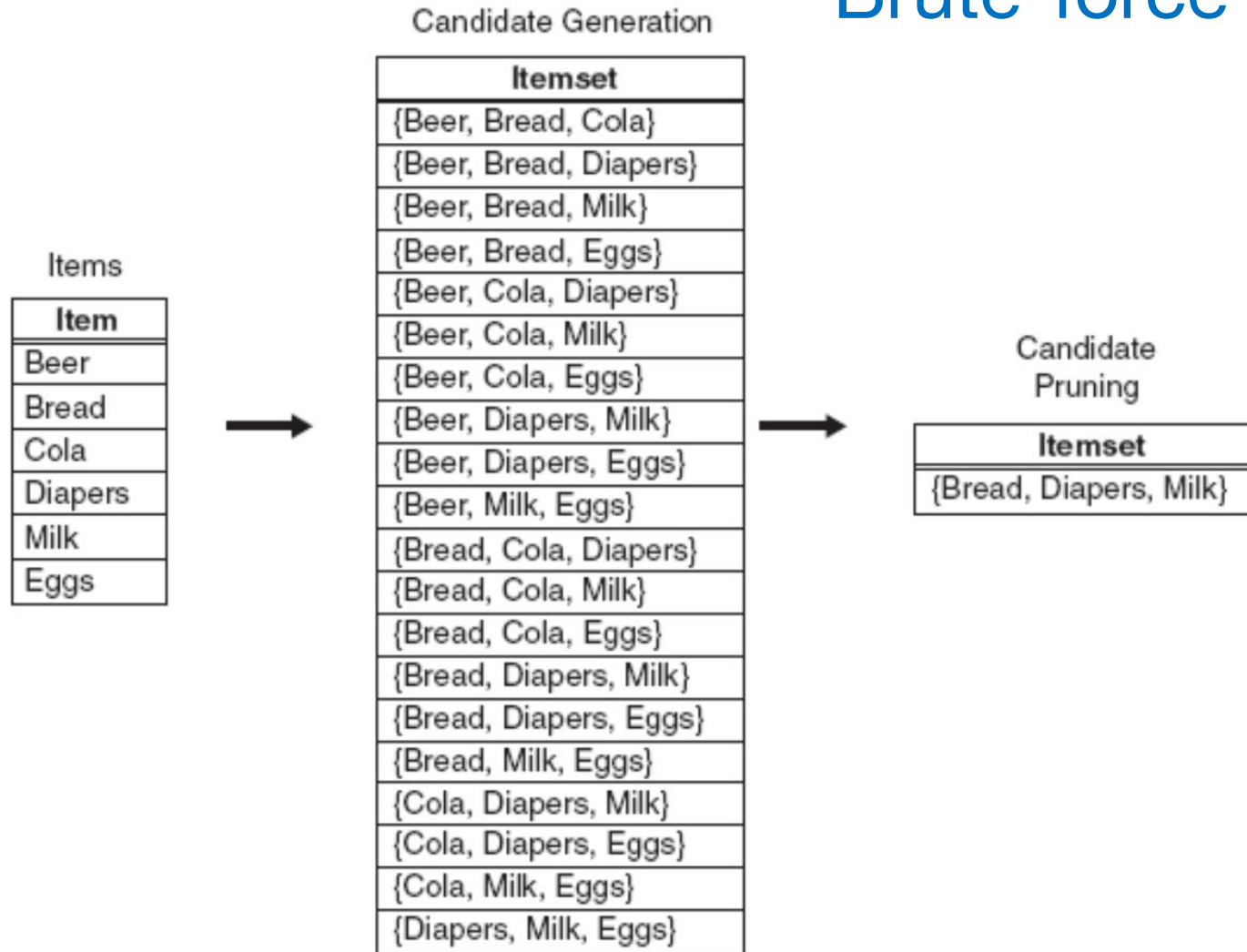


Candidate Generation

- Three Strategies
 - Brute-force approach
 - $L_k \times L_1 \rightarrow C_{k+1}$
 - $L_k \times L_K \rightarrow C_{k+1}$

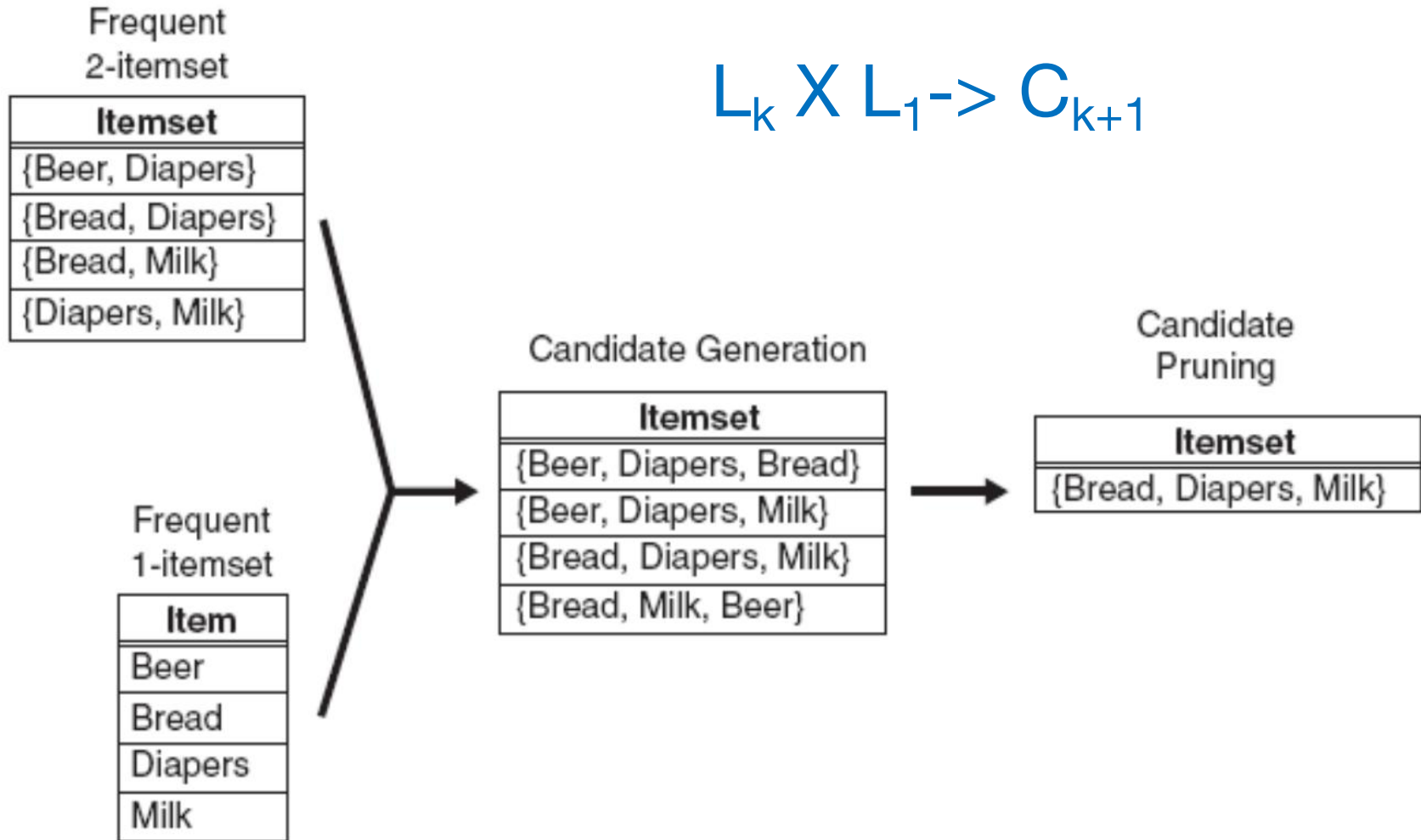
Candidate Generation

Brute-force approach

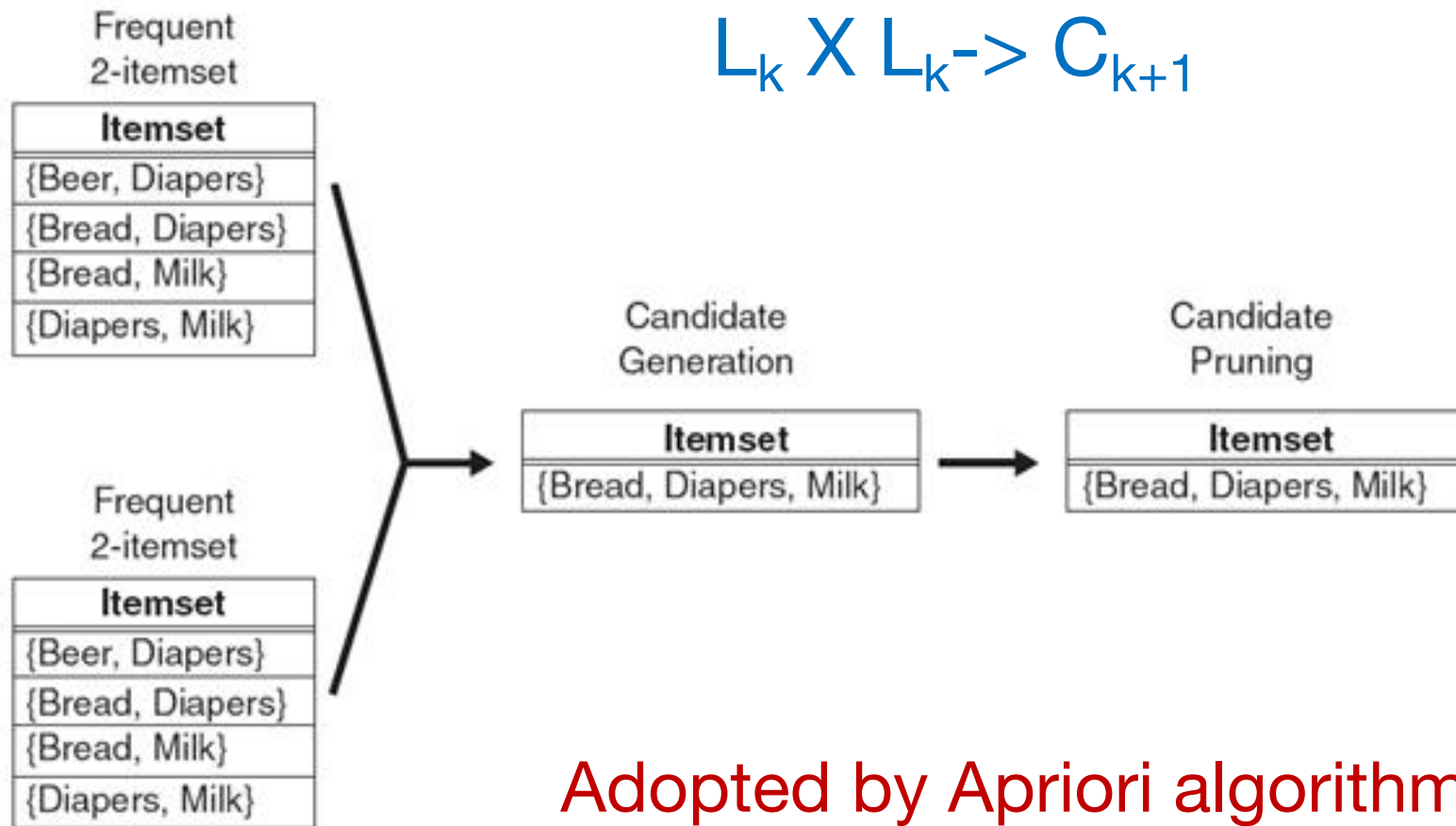


Candidate Generation

$$L_k \times L_1 \rightarrow C_{k+1}$$

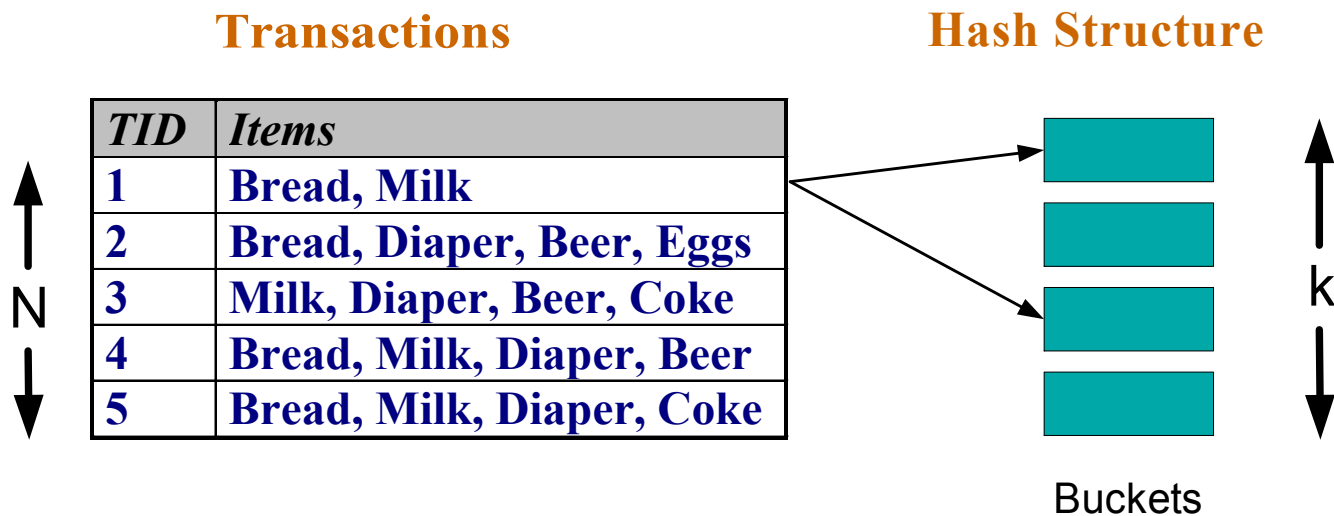


Candidate Generation



Reducing Number of Comparisons

- Candidate counting:
 - Scan the database of transactions to determine the support of each candidate itemset
 - To reduce the number of comparisons, store the candidates in a hash structure
 - Instead of matching each transaction against every candidate, match it against candidates contained in the hashed buckets



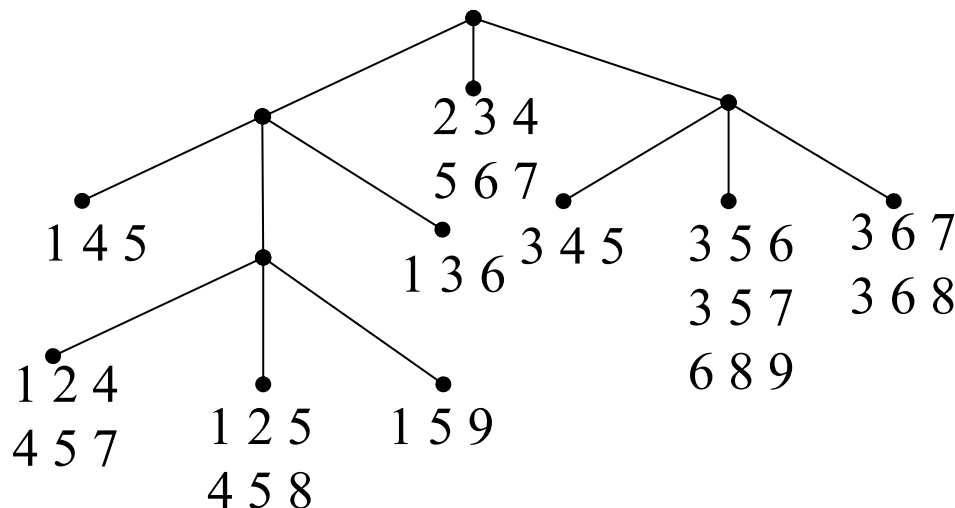
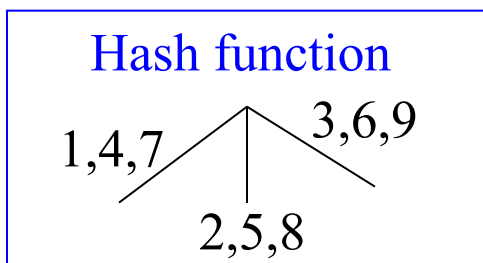
Generate Hash Tree

Suppose you have 15 candidate itemsets of length 3:

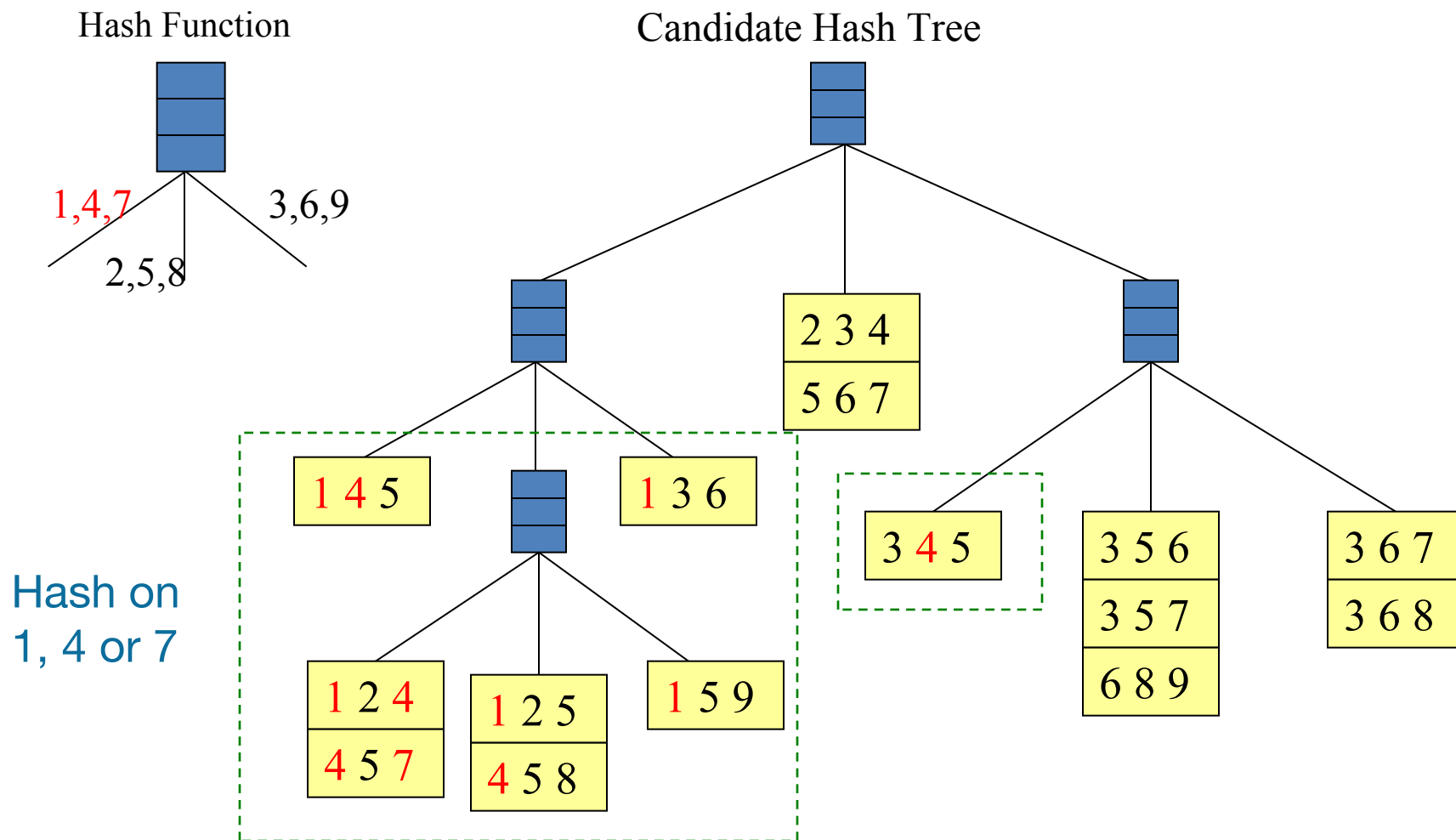
{1 4 5}, {1 2 4}, {4 5 7}, {1 2 5}, {4 5 8}, {1 5 9}, {1 3 6}, {2 3 4}, {5 6 7}, {3 4 5}, {3 5 6}, {3 5 7}, {6 8 9}, {3 6 7}, {3 6 8}

You need:

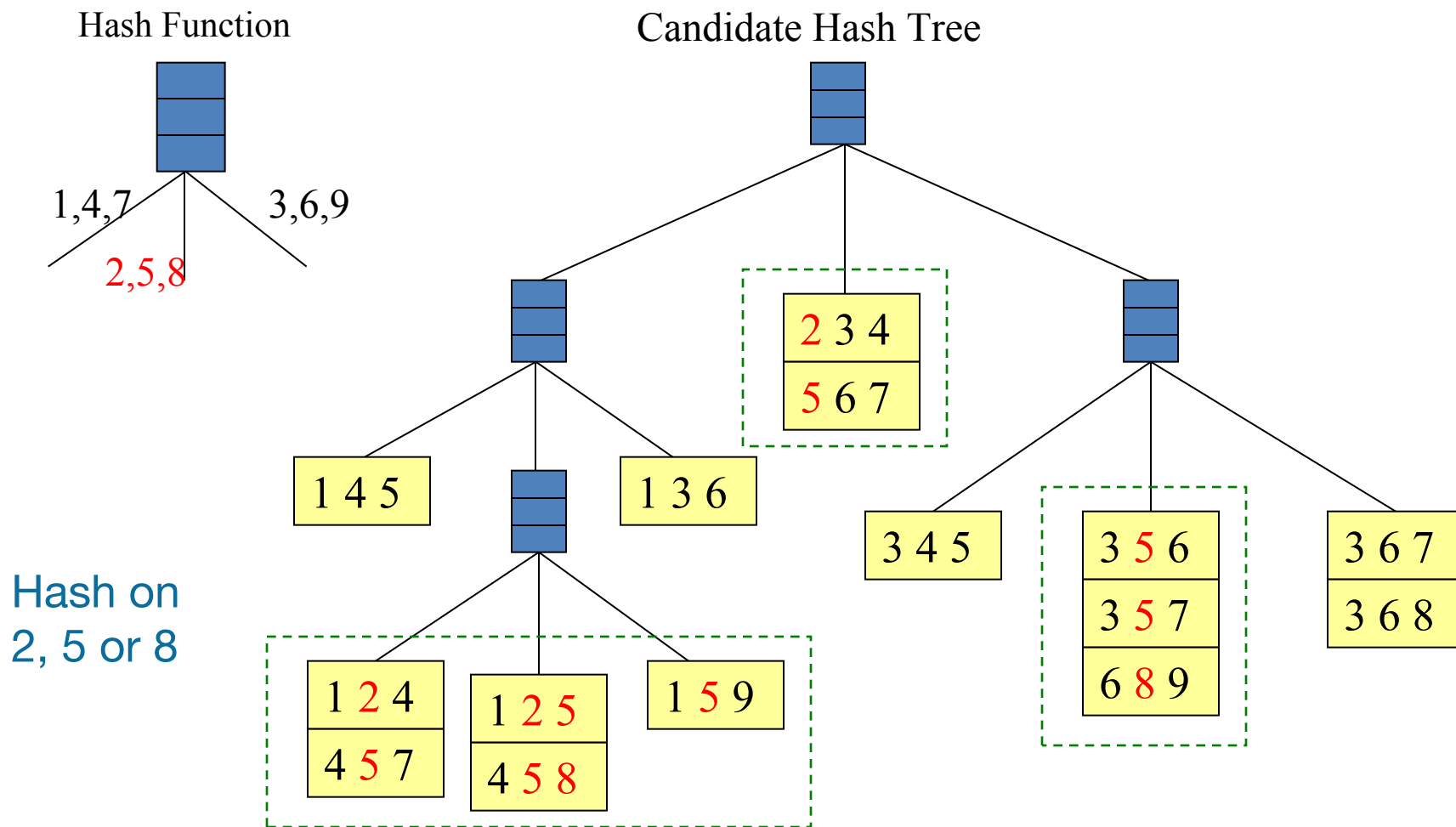
- Hash function
- Max leaf size: max number of itemsets stored in a leaf node (if number of candidate itemsets exceeds max leaf size, split the node)



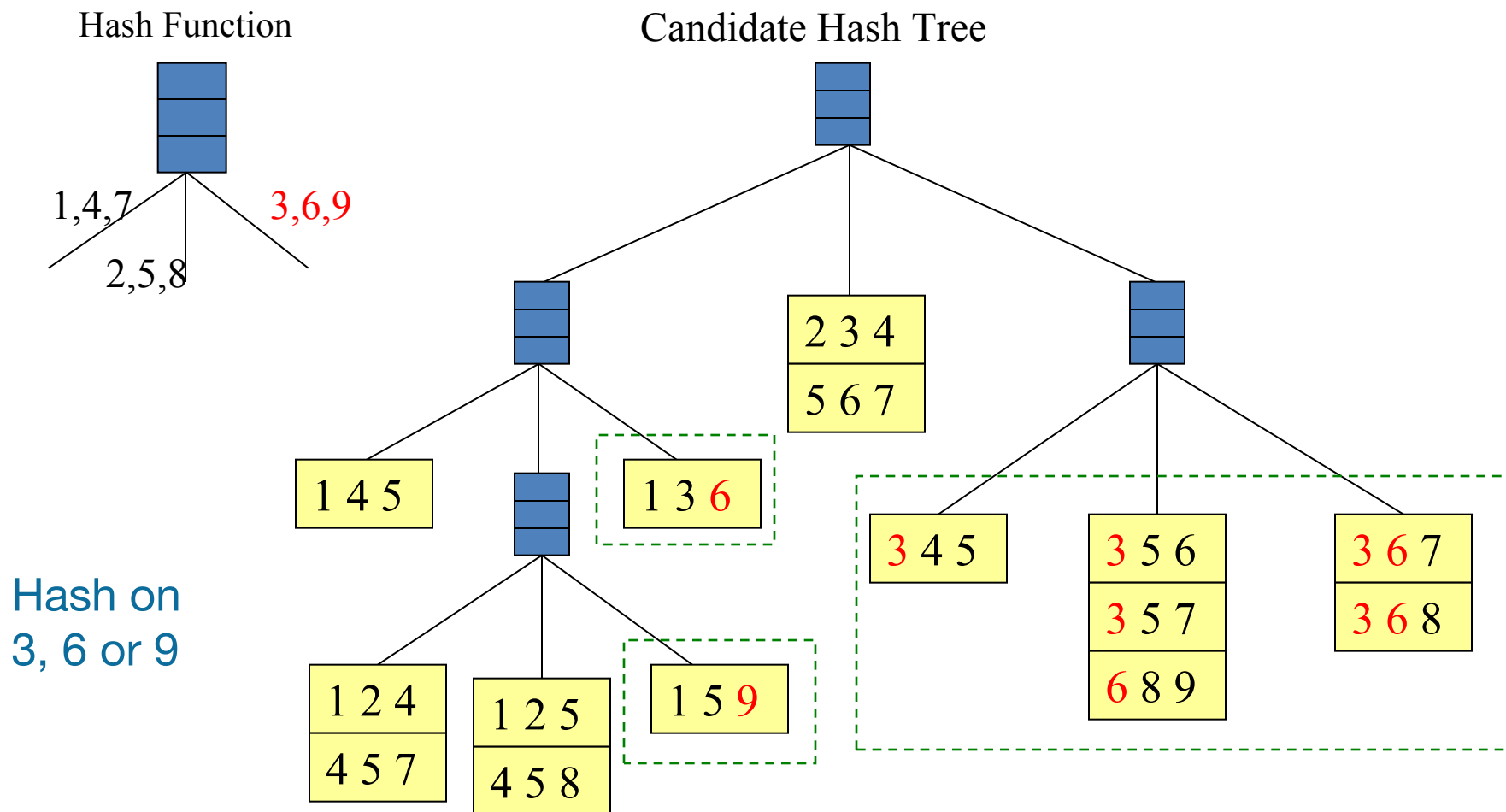
Association Rule Discovery: Hash tree



Association Rule Discovery: Hash tree

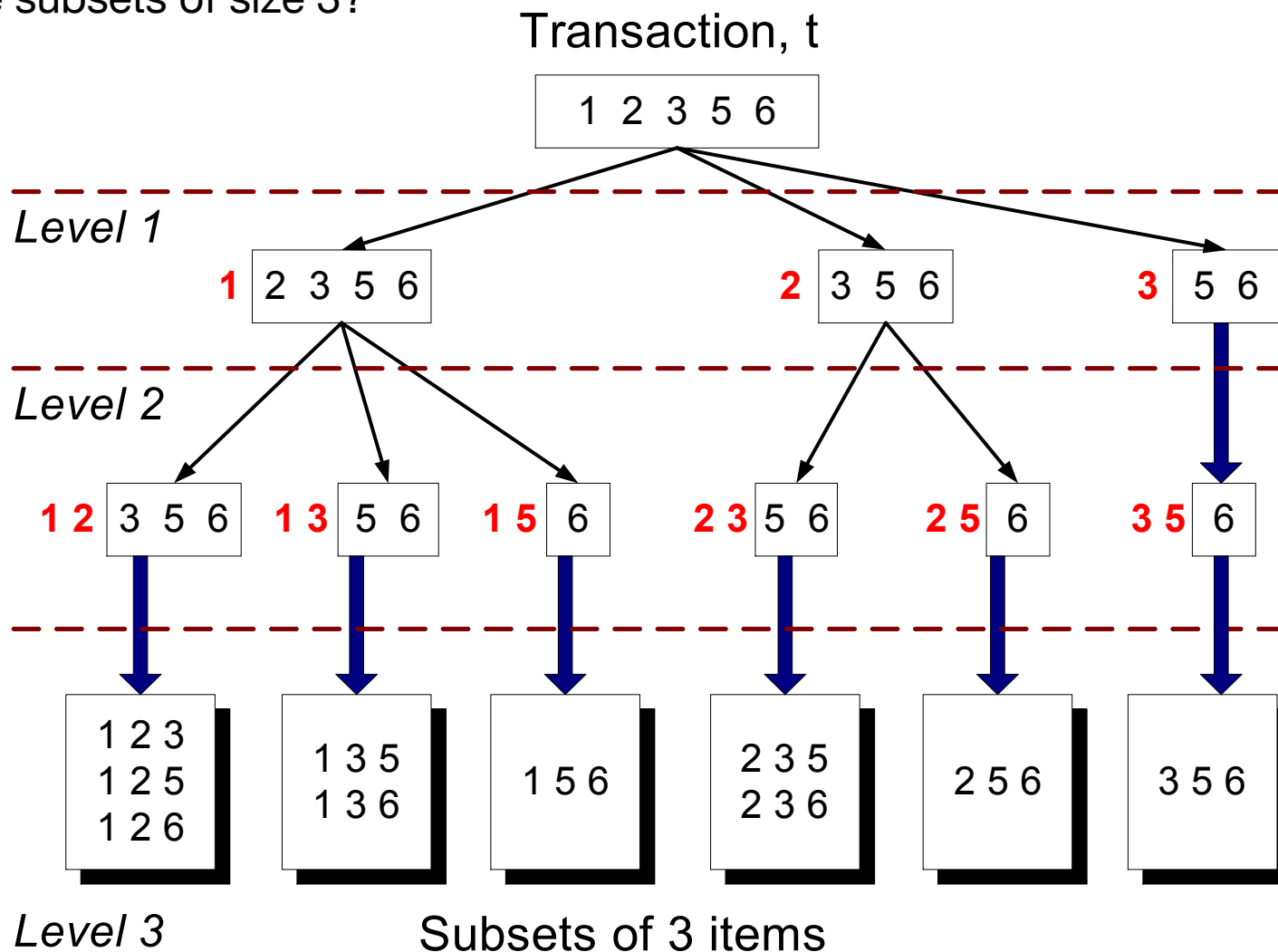


Association Rule Discovery: Hash tree

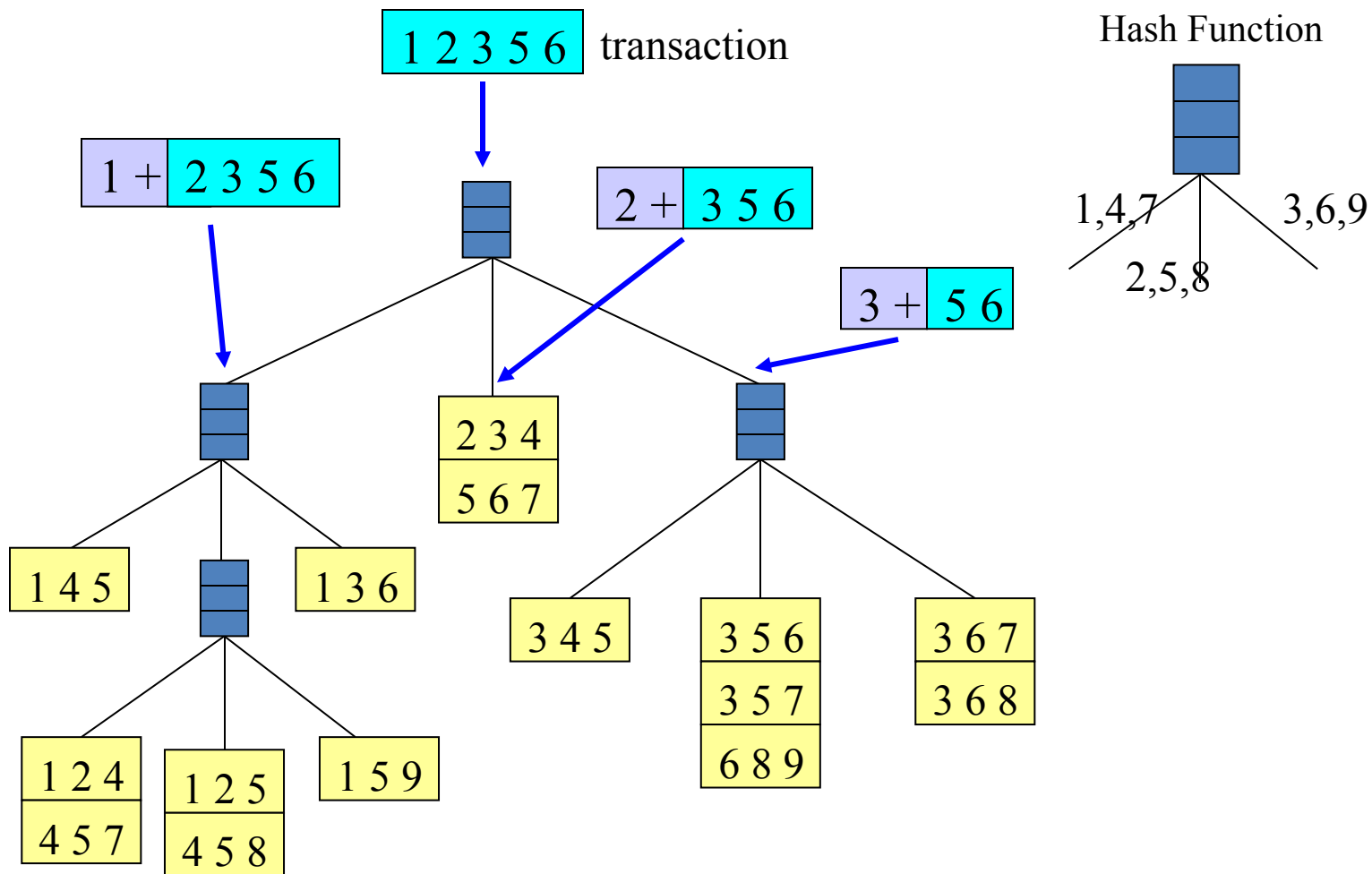


Subset Operation

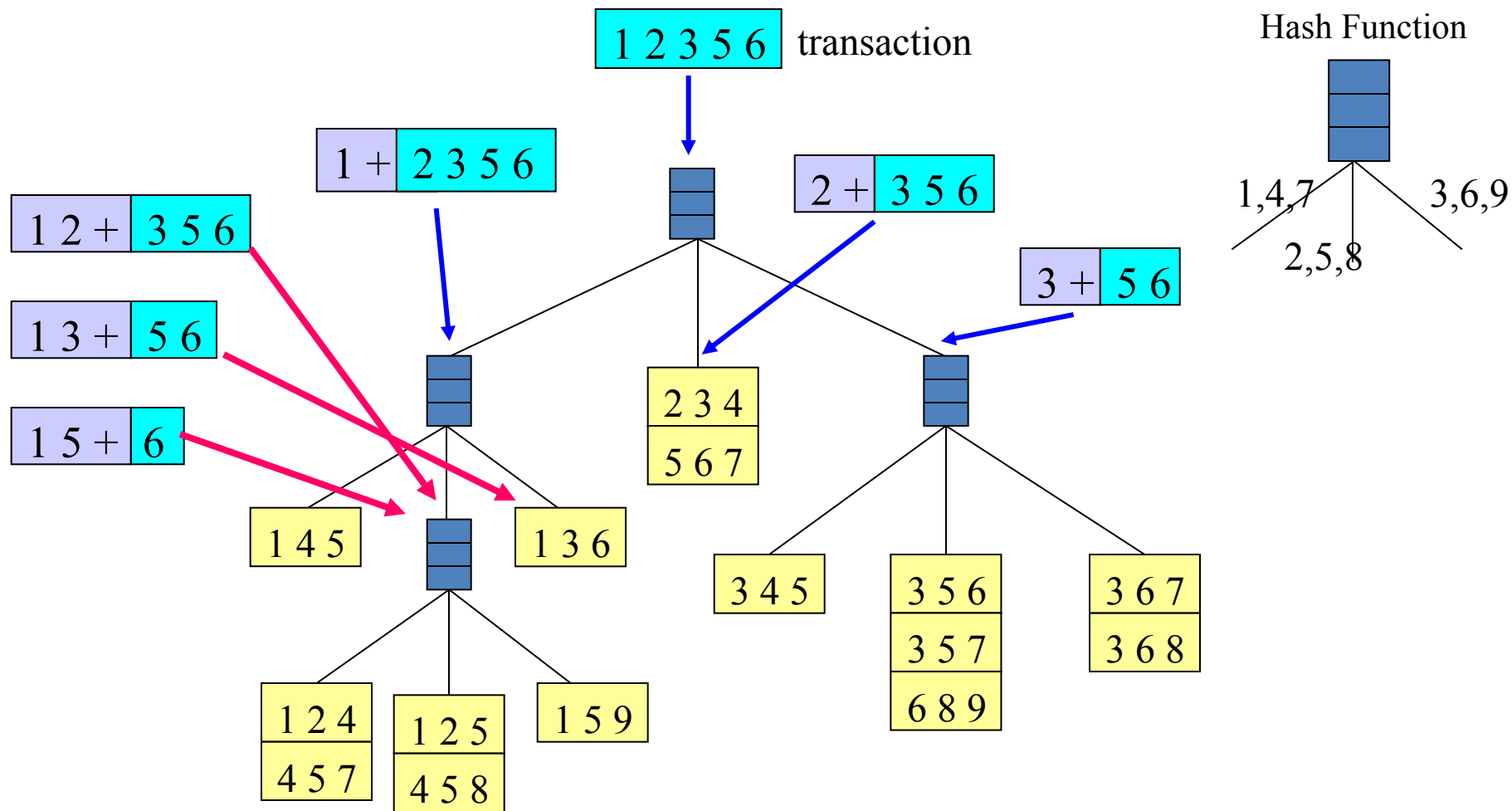
Given a transaction t , what are the possible subsets of size 3?



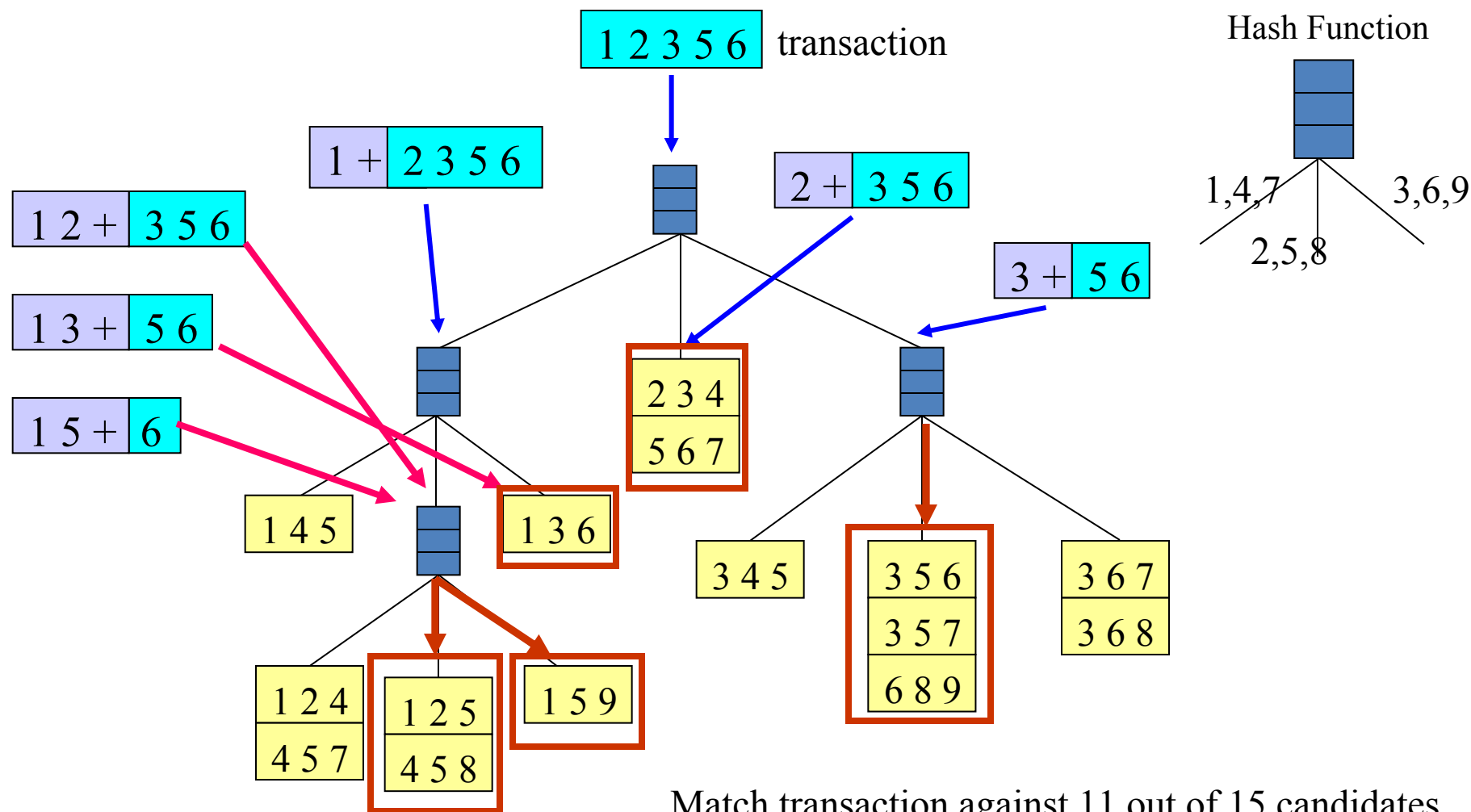
Subset Operation Using Hash Tree



Subset Operation Using Hash Tree



Subset Operation Using Hash Tree



Factors Affecting Complexity

- Choice of minimum support threshold
 - lowering support threshold results in more frequent itemsets
 - this may increase number of candidates and max length of frequent itemsets
- Dimensionality (number of items) of the data set
 - more space is needed to store support count of each item
 - if number of frequent items also increases, both computation and I/O costs may also increase
- Size of database
 - since Apriori makes multiple passes, run time of algorithm may increase with number of transactions
- Average transaction width
 - transaction width increases with denser data sets
 - This may increase max length of frequent itemsets and traversals of hash tree (number of subsets in a transaction increases with its width)



Mining Association Rules

- Two-step approach:
 1. Frequent Itemset Generation
 - Generate all itemsets whose support $\geq$ minsup
 2. Rule Generation
 - Generate high confidence rules from each frequent itemset, where each rule is a binary partitioning of a frequent itemset

Rule Generation

- Given a frequent itemset L , find all non-empty subsets $f \subset L$ such that $f \rightarrow L - f$ satisfies the minimum confidence requirement
 - If $\{A,B,C,D\}$ is a frequent itemset, candidate rules:

$ABC \rightarrow D,$	$ABD \rightarrow C,$	$ACD \rightarrow B,$	$BCD \rightarrow A,$
$A \rightarrow BCD,$	$B \rightarrow ACD,$	$C \rightarrow ABD,$	$D \rightarrow ABC$
$AB \rightarrow CD,$	$AC \rightarrow BD,$	$AD \rightarrow BC,$	$BC \rightarrow AD,$
$BD \rightarrow AC,$	$CD \rightarrow AB,$		
- If $|L| = k$, then there are $2^k - 2$ candidate association rules (ignoring $L \rightarrow \emptyset$ and $\emptyset \rightarrow L$)

Rule Generation

- How to efficiently generate rules from frequent itemsets?

- In general, confidence does not have an anti-monotone property

- $c(ABC \rightarrow D)$ can be larger or smaller than $c(AB \rightarrow D)$

- But confidence of rules generated from the same itemset has an anti-monotone property

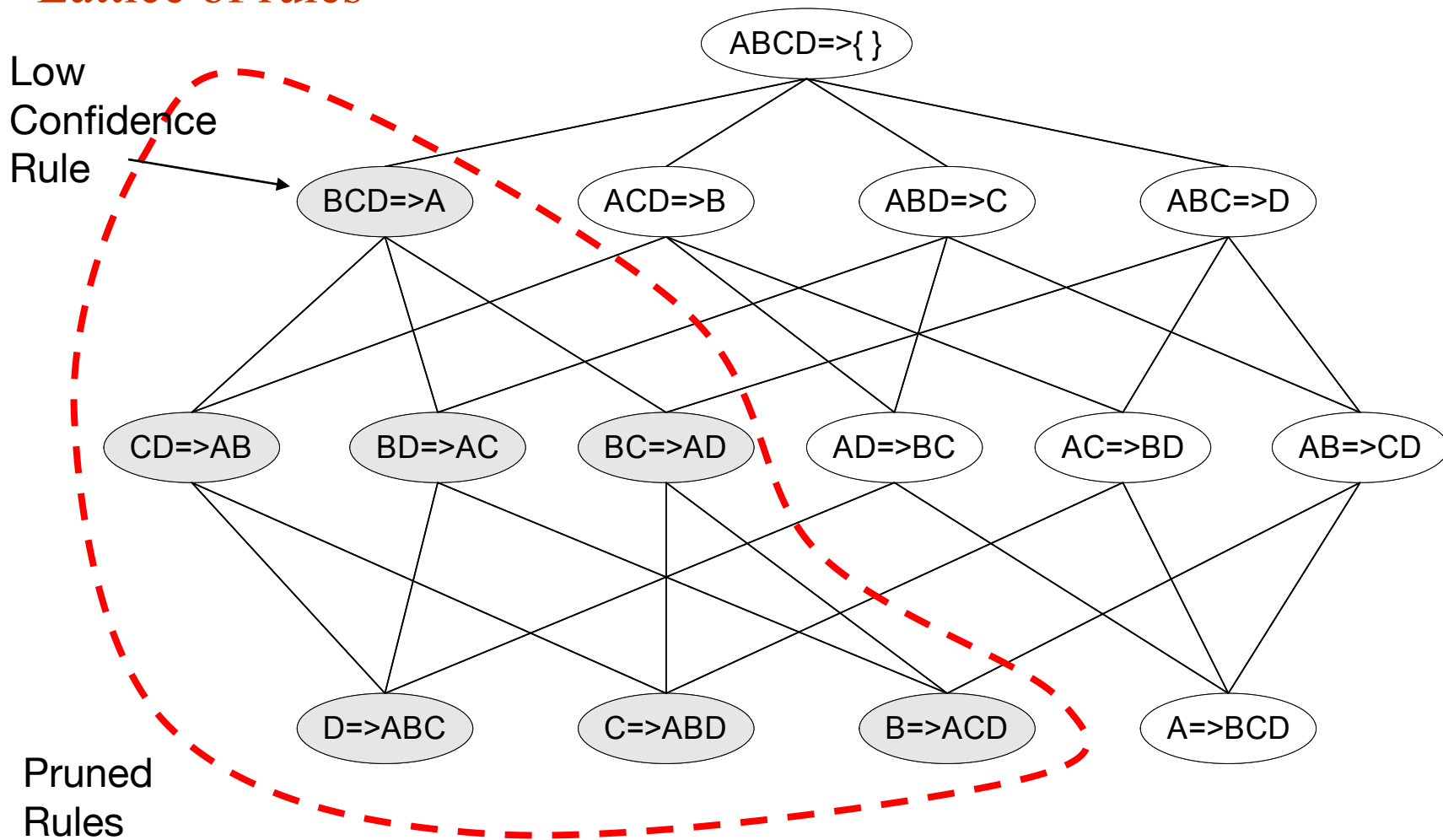
- e.g., $L = \{A, B, C, D\}$:

$$c(ABC \rightarrow D) \geq c(AB \rightarrow CD) \geq c(A \rightarrow BCD)$$

- Confidence is anti-monotone w.r.t. number of items on the RHS of the rule

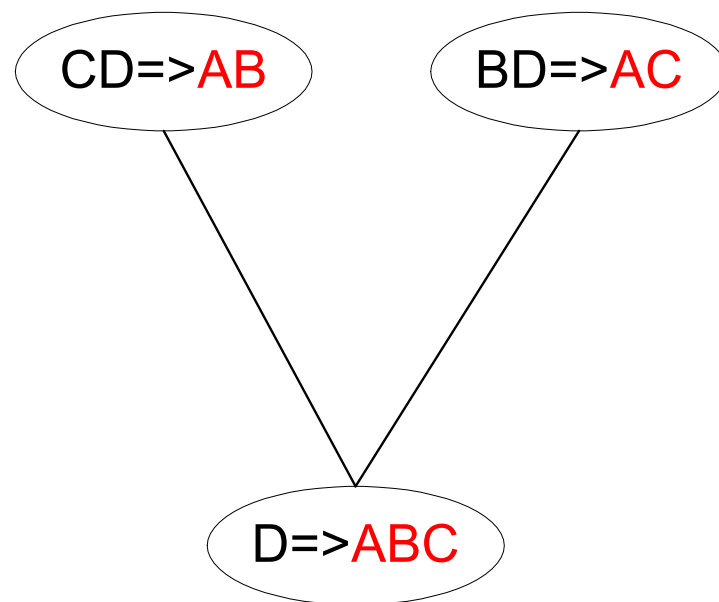
Rule Generation for Apriori Algorithm

Lattice of rules



Rule Generation for Apriori Algorithm

- Candidate rule is generated by merging two rules that share the same prefix in the rule consequent
- $\text{join}(\text{CD} \Rightarrow \text{AB}, \text{BD} \Rightarrow \text{AC})$ would produce the candidate rule $\text{D} \Rightarrow \text{ABC}$
- Prune rule $\text{D} \Rightarrow \text{ABC}$ if its subset $\text{AD} \Rightarrow \text{BC}$ does not have high confidence



- Basic Concepts
 - Item, Confidence, Support
- Apriori Algorithm
 - Frequent Item generation
 - Rule generation
- Pattern Evaluation

Pattern Evaluation

- Association rule algorithms tend to produce too many rules
 - many of them are uninteresting or redundant
 - Redundant if $\{A,B,C\} \rightarrow \{D\}$ and $\{A,B\} \rightarrow \{D\}$ have same support & confidence
- Interestingness measures can be used to prune/rank the derived patterns
- In the original formulation of association rules, support & confidence are the only measures used

Computing Interestingness Measure

- Given a rule $X \rightarrow Y$, information needed to compute rule interestingness can be obtained from a contingency table

Contingency table for $X \rightarrow Y$

	Y	$\overline{Y}$	
X	f_{11}	f_{10}	f_{1+}
$\overline{X}$	f_{01}	f_{00}	f_{0+}
	f_{+1}	f_{+0}	$ T $

f_{11} : support of X and Y

f_{10} : support of $\overline{X}$ and Y

f_{01} : support of X and $\overline{Y}$

f_{00} : support of $\overline{X}$ and $\overline{Y}$

Used to define various measures

support, confidence, lift, Gini, J-measure, etc.

Drawback of Confidence

	Coffee	<u>Coffee</u>	
Tea	15	5	20
<u>Tea</u>	75	5	80
	90	10	100

Association Rule: Tea $\rightarrow$ Coffee

Confidence = $P(\text{Coffee}|\text{Tea}) = 0.75$

but $P(\text{Coffee}) = 0.9$

$\Rightarrow$ Although confidence is high, rule is misleading

$\Rightarrow P(\text{Coffee}|\overline{\text{Tea}}) = 0.9375$

Statistical Independence

- Population of 1000 students
 - 600 students know how to swim (S)
 - 700 students know how to bike (B)
 - 420 students know how to swim and bike (S,B)
 - $P(S \cap B) = 420/1000 = 0.42$
 - $P(S) \times P(B) = 0.6 \times 0.7 = 0.42$
 - $P(S \cap B) = P(S) \times P(B) \Rightarrow$ Statistical independence
 - $P(S \cap B) > P(S) \times P(B) \Rightarrow$ Positively correlated
 - $P(S \cap B) < P(S) \times P(B) \Rightarrow$ Negatively correlated

Statistical-based Measures

- Measures that take into account statistical dependence

$$\textit{Lift} = \frac{P(Y | X)}{P(Y)}$$

$$\textit{Interest} = \frac{P(X, Y)}{P(X)P(Y)}$$

$$PS = P(X, Y) - P(X)P(Y)$$

$$\phi - \textit{coefficient} = \frac{P(X, Y) - P(X)P(Y)}{\sqrt{P(X)[1 - P(X)]P(Y)[1 - P(Y)]}}$$

Example: Lift/Interest

	Coffee	<u>Coffee</u>	
Tea	15	5	20
<u>Tea</u>	75	5	80
	90	10	100

Association Rule: Tea $\rightarrow$ Coffee

Confidence= $P(\text{Coffee}|\text{Tea}) = 0.75$

but $P(\text{Coffee}) = 0.9$

$\Rightarrow \text{Lift} = 0.75/0.9 = 0.8333 (< 1, \text{ therefore is negatively associated})$

Measures

There are lots of measures proposed in the literature

Some measures are good for certain applications, but not for others

#	Measure	Formula
1	ϕ -coefficient	$\frac{P(A,B) - P(A)P(B)}{\sqrt{P(A)P(B)(1-P(A))(1-P(B))}}$
2	Goodman-Kruskal's (λ)	$\frac{\sum_j \max_k P(A_j, B_k) + \sum_k \max_j P(A_j, B_k) - \max_j P(A_j) - \max_k P(B_k)}{2 - \max_j P(A_j) - \max_k P(B_k)}$
3	Odds ratio (α)	$\frac{P(A,B)P(\bar{A},\bar{B})}{P(A,\bar{B})P(\bar{A},B)}$
4	Yule's Q	$\frac{P(A,B)P(\bar{A}\bar{B}) - P(A,\bar{B})P(\bar{A},B)}{P(A,B)P(\bar{A}\bar{B}) + P(A,\bar{B})P(\bar{A},B)} = \frac{\alpha-1}{\alpha+1}$
5	Yule's Y	$\frac{\sqrt{P(A,B)P(\bar{A}\bar{B})} - \sqrt{P(A,\bar{B})P(\bar{A},B)}}{\sqrt{P(A,B)P(\bar{A}\bar{B})} + \sqrt{P(A,\bar{B})P(\bar{A},B)}} = \frac{\sqrt{\alpha-1}}{\sqrt{\alpha+1}}$
6	Kappa (κ)	$\frac{P(A,B) + P(\bar{A},\bar{B}) - P(A)P(B) - P(\bar{A})P(\bar{B})}{1 - P(A)P(B) - P(\bar{A})P(\bar{B})}$
7	Mutual Information (M)	$\frac{\sum_i \sum_j P(A_i, B_j) \log \frac{P(A_i, B_j)}{P(A_i)P(B_j)}}{\min(-\sum_i P(A_i) \log P(A_i), -\sum_j P(B_j) \log P(B_j))}$
8	J-Measure (J)	$\max \left(P(A, B) \log \left(\frac{P(B A)}{P(B)} \right) + P(\bar{A}\bar{B}) \log \left(\frac{P(\bar{B} \bar{A})}{P(\bar{B})} \right), \right.$ $\left. P(A, B) \log \left(\frac{P(A B)}{P(A)} \right) + P(\bar{A}\bar{B}) \log \left(\frac{P(\bar{A} \bar{B})}{P(\bar{A})} \right) \right)$
9	Gini index (G)	$\max \left(P(A)[P(B A)^2 + P(\bar{B} A)^2] + P(\bar{A})[P(B \bar{A})^2 + P(\bar{B} \bar{A})^2] \right.$ $\left. - P(B)^2 - P(\bar{B})^2, \right.$ $\left. P(B)[P(A B)^2 + P(\bar{A} B)^2] + P(\bar{B})[P(A \bar{B})^2 + P(\bar{A} \bar{B})^2] \right.$ $\left. - P(A)^2 - P(\bar{A})^2 \right)$
10	Support (s)	$P(A, B)$
11	Confidence (c)	$\max(P(B A), P(A B))$
12	Laplace (L)	$\max \left(\frac{NP(A,B)+1}{NP(A)+2}, \frac{NP(A,B)+1}{NP(B)+2} \right)$
13	Conviction (V)	$\max \left(\frac{P(A)P(\bar{B})}{P(\bar{A}B)}, \frac{P(B)P(\bar{A})}{P(\bar{B}A)} \right)$
14	Interest (I)	$\frac{P(A,B)}{P(A)P(B)}$
15	cosine (IS)	$\frac{P(A,B)}{\sqrt{P(A)P(B)}}$
16	Piatetsky-Shapiro's (PS)	$P(A, B) - P(A)P(B)$
17	Certainty factor (F)	$\max \left(\frac{P(B A) - P(B)}{1 - P(B)}, \frac{P(A B) - P(A)}{1 - P(A)} \right)$
18	Added Value (AV)	$\max(P(B A) - P(B), P(A B) - P(A))$
19	Collective strength (S)	$\frac{P(A,B) + P(\bar{A}\bar{B})}{P(A)P(B) + P(\bar{A})P(\bar{B})} \times \frac{1 - P(A)P(B) - P(\bar{A})P(\bar{B})}{1 - P(A,B) - P(\bar{A}\bar{B})}$
20	Jaccard (ζ)	$\frac{P(A,B)}{P(A) + P(B) - P(A,B)}$
21	Klogsen (K)	$\sqrt{P(\bar{A},\bar{B})} \max(P(B A) - P(B), P(A B) - P(A))$

Simpson's Paradox

Contingency table

	Fitness Equip.=YES	Fitness Equip.=NO	Total
HDTV=YES	99	81	180
HDTV=NO	54	66	120
	153	147	300

- Rule 1: (HDTV=YES) $\rightarrow$ (Fitness Equip.=YES)
- Rule 2: (HDTV=NO) $\rightarrow$ (Fitness Equip.=YES)

What is the confidence value of those two rules?

Confidence (Rule 1) = ?

Confidence (Rule 2) = ?

Simpson's Paradox

Contingency table

		Fitness Equip.=YES	Fitness Equip.=NO	Total
College Students	HDTV=YES	1	9	10
	HDTV=NO	4	30	34
white-collar worker	HDTV=YES	98	72	170
	HDTV=NO	50	36	86

- Rule 1: (HDTV=YES) $\rightarrow$ (Fitness Equip.=YES)
- Rule 2: (HDTV=NO) $\rightarrow$ (Fitness Equip.=YES)

What is the confidence value of those two rules?

Conf.(Rule1|college student)= ?

Conf.(Rule2|college student)= ?

Conf.(Rule1|white-collar worker)= ?

Conf.(Rule2|white-collar worker)= ?

Case Study on Groceries Data

• Data Information

shrimp	almonds	avocado	vegetables mix	green grapes	whole weat flour	yams
burgers	meatballs	eggs				
chutney						
turkey	avocado					
mineral water	milk	energy bar	whole wheat rice	green tea		
low fat yogurt						
whole wheat pasta	french fries					
soup	light cream	shallot				
frozen vegetables	spaghetti	green tea				
french fries						
eggs	pet food					
cookies						
turkey	burgers	mineral water	eggs	cooking oil		
spaghetti	champagne	cookies				
mineral water	salmon					
mineral water						
shrimp	chocolate	chicken	honey	oil	cooking oil	low fat yogurt
turkey	eggs					
turkey	fresh tuna	tomatoes	spaghetti	mineral water	black tea	salmon
meatballs	milk	honey	french fries	protein bar		
red wine	shrimp	pasta	pepper	eggs	chocolate	shampoo
rice	sparkling water					
spaghetti	mineral water	ham	body spray	pancakes	green tea	
burgers	grated cheese	shrimp	pasta	avocado	honey	white wine
eggs						
parmesan cheese	spaghetti	soup	avocado	milk	fresh bread	
ground beef	spaghetti	mineral water	milk	energy bar	black tea	salmon

Association Rule

```
In [24]: # 生成频繁项集
freq = apriori(df, min_support = 0.05, use_colnames = True)
```

```
In [25]: freq
```

```
Out[25]:
```

	support	itemsets
0	0.087188	(burgers)
1	0.081056	(cake)
2	0.059992	(chicken)
3	0.163845	(chocolate)
4	0.080389	(cookies)
5	0.051060	(cooking oil)
6	0.179709	(eggs)
7	0.079323	(escalope)
8	0.170911	(french fries)
9	0.063325	(frozen smoothie)
10	0.095321	(frozen vegetables)
11	0.052393	(grated cheese)

```
In [26]: # 计算关联规则
Rules = association_rules(freq, metric = "confidence", min_threshold = 0.2)
Rules
```

```
Out[26]:
```

	antecedents	consequents	antecedent support	consequent support	support	confidence	lift	leverage	conviction
0	(mineral water)	(chocolate)	0.238368	0.163845	0.052660	0.220917	1.348332	0.013604	1.073256
1	(chocolate)	(mineral water)	0.163845	0.238368	0.052660	0.321400	1.348332	0.013604	1.122357
2	(mineral water)	(eggs)	0.238368	0.179709	0.050927	0.213647	1.188845	0.008090	1.043158
3	(eggs)	(mineral water)	0.179709	0.238368	0.050927	0.283383	1.188845	0.008090	1.062815
4	(mineral water)	(spaghetti)	0.238368	0.174110	0.059725	0.250559	1.439085	0.018223	1.102008
5	(spaghetti)	(mineral water)	0.174110	0.238368	0.059725	0.343032	1.439085	0.018223	1.159314



Some Takeaways

- Association Rule
- Apriori Algorithm
- Pattern Evaluation



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- Introduction to data mining (by Tan, Steinbach, Kumar)
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The End

■ THANK YOU

Any questions?

